Two Robot Arms That Build With LEGO

How the project works, explained in simple words

ISDN Robotic LEGO Assembly project

November 2026

What this report is. A plain-language tour of the project: how a drawing becomes a list of bricks, how the robots decide the order and when one needs to hold the model steady, how the computer pretends to be the real world, and how each arm picks up, carries and clicks in every brick without bumping into anything. Each part has a flowchart: start at the green box, follow the arrows, answer the yellow questions with "yes" or "no", and stop at the red box.

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1 The big picture

Imagine you want two robot arms to build a LEGO model for you. You draw the model from the front and from the side. The computer then has to answer four questions:

1. **What bricks make this shape?** (the *blueprint*)
2. **In what order should they go on?** (the *plan*)
3. **Will the model wobble or break while a brick is pushed on?** If so, the other arm holds it steady (a *support*).
4. **How does each arm actually do it?** Pick up, carry, place, check, and never crash into the other arm (the *control*).

Everything happens inside a computer *simulation*: a pretend world where bricks fall, slide and click like real ones. Two robot arms (called A and B) each build their own model at the same time, side by side, and help each other when one model needs holding.

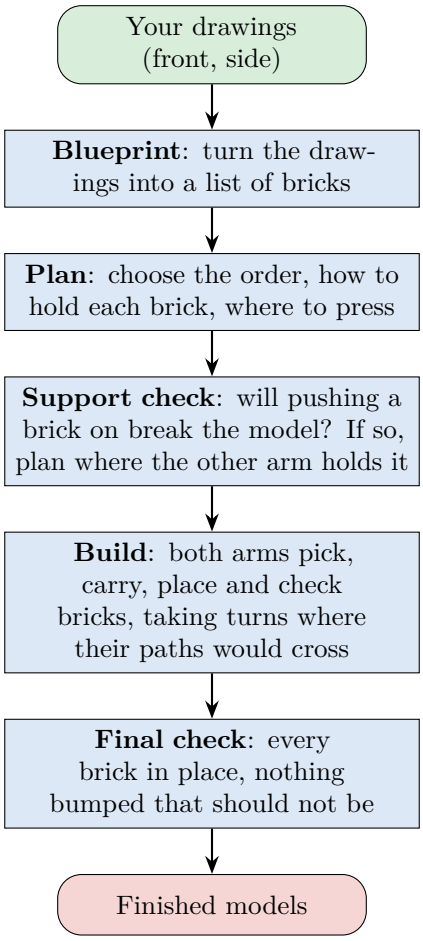


Figure 1: The whole project, from drawing to finished model.

2 Making the blueprint: from drawing to bricks

2.1 The idea in simple words

LEGO bricks sit on an invisible 3D grid. One step sideways is one stud (8 mm) and one step up is one brick height (9.6 mm). So we first build the model out of tiny grid cubes, and then cover those

cubes with real bricks.

Step 1 – Read the drawings. Each drawing is a dark shape on a light background (or the opposite). The computer finds the shape and cuts away the empty border.

Step 2 – Fit the drawing onto the grid. You say how many studs wide the model is. From that, the computer works out how many layers tall it must be, using the real stud and brick sizes. Each grid square is marked "full" if more than half of it is dark in the drawing.

Step 3 – Carve the model, like a shadow puzzle. Start with a solid block of grid cubes. Look at it from the front: any cube outside the front shape is removed. Look from the side: any cube outside the side shape is removed. What is left matches every drawing. (A hidden hole that no drawing can see is simply filled in.)

Step 4 – Cover the cubes with bricks, one layer at a time, from the bottom. The computer can use these bricks: 2×6, 2×4, 2×3, 2×2, 1×6, 1×4, 1×2 and 1×1. In each layer:

- Cells that hang over empty space are covered first, by a brick that sits on as many bricks below as possible. That is what ties the two legs of an arch together.
- Every brick must sit on at least one brick below. If a cell cannot be reached that way, the shape overhangs too far and the computer says so.
- When there is a choice, it prefers a brick that does not put a joint right on top of a joint below (like a brick wall), then bigger bricks, then bricks that point the other way from the layer below (crossing layers makes the model stronger).

Step 5 – Number the bricks in the order they will be built.

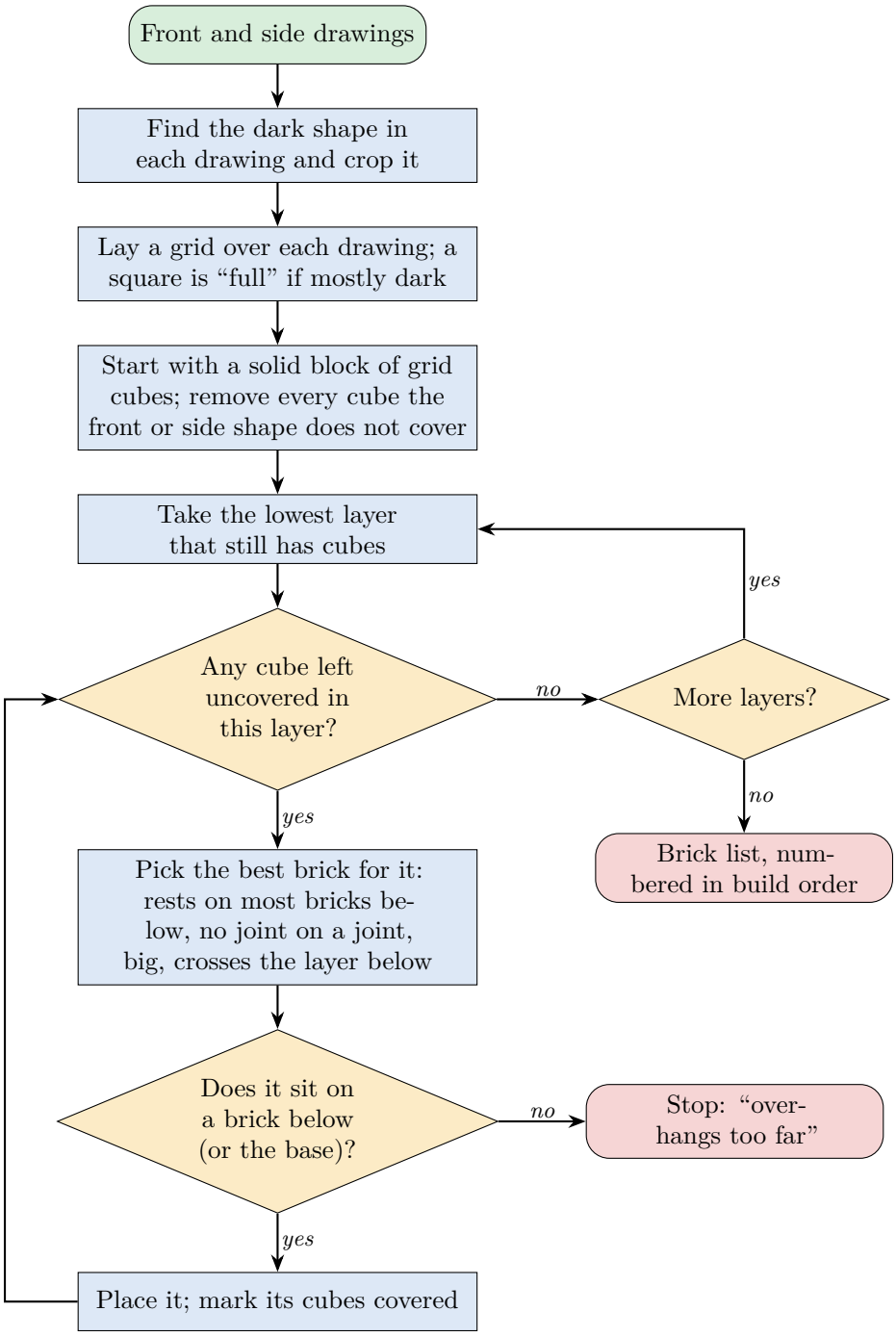


Figure 2: Turning drawings into a list of bricks.

3 Making the plan: order, grip and press

The blueprint says *which* bricks. The plan says *how* to build them. For every brick the plan writes down one “step”:

- **When.** Bricks go on from the bottom layer up. A brick may only be placed once the bricks under it are there. Inside a layer, square 2×2 bricks go first, so the fingers that hold them do not rub against a neighbour.
- **Where.** The exact spot and turn of the brick on the base plate, and a spot just above it (“hover point”) where the arm stops before going down.

- **How to hold it.** The gripper has two flat fingers that squeeze two opposite sides of the brick. The plan picks the pair of sides where the fingers touch the fewest bricks already standing next to it, and how wide to open.
- **Where to press.** If only part of a brick sits on bricks below (an overhang, like a shelf), pressing it in the middle could tip it. So the plan slides the grip toward the part that is supported, choosing the spot that puts the least strain on the model.

Tight spots. Sometimes a brick's neighbours are so close that the fingers cannot reach down beside it. Two things help:

- **Reorder small bricks.** If a tiny 1×1 brick would end up boxed in, its layer is rebuilt smallest-bricks-first, so the 1×1 goes in while there is still room for the fingers. (A dropped 1×1 rolls off its single stud and falls over, so it must be held all the way down.)
- **Grip fallbacks.** The arm tries grips in a fixed order: the planned one, the other pair of sides, then holding the brick a little off-centre, nearest first. If none fits, it uses *drop-and-press*: let go of the brick just above its spot, then push it down with the closed fingertips.

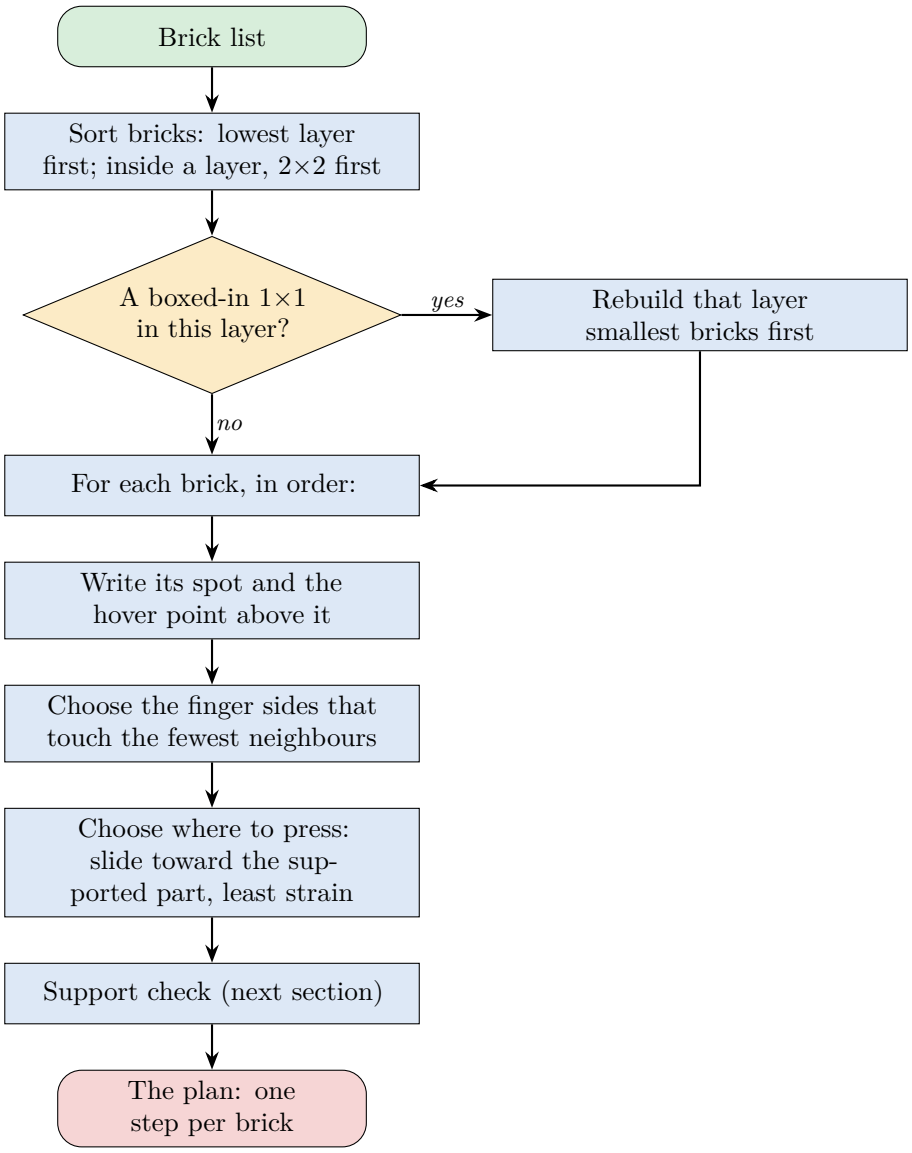


Figure 3: Turning a brick list into a step-by-step plan.

4 Does the model need a helping hand?

4.1 Why a model can break while you build it

LEGO bricks hold together because the studs squeeze into the brick above. Each stud holds about 11.3 newtons before it pops off (a newton is roughly the weight of a small apple). Pushing a new brick on needs about 8.9 newtons per stud. So when the arm presses a brick onto a tall or overhanging part, that push can lever another joint apart, like pressing on the end of a diving board.

4.2 How the computer checks

For every step, the computer imagines the press. It uses 1.3 times the normal push, to be on the safe side. Then it works out how hard every joint in the model is being pulled or bent. Each joint gets a score: its load divided by its strength. A score of 1 means "right at breaking point".

- It multiplies the score by a safety factor of 1.5. If the result is 1 or more, that step **needs support**.
- Then it finds the **weakest joint**, the one closest to breaking.
- It tries ways for the other arm to grip the model near that joint. The helper's fingers squeeze the bricks on both sides of the weak joint, so the push goes into the robot instead of through the joint. It keeps the grip that gives the lowest score.
- It checks that the two hands fit: the helper leans its hand away (45, 60 or 75 degrees) so it does not get in the way of the placing hand.
- If no grip makes the step safe, planning stops with a clear message.

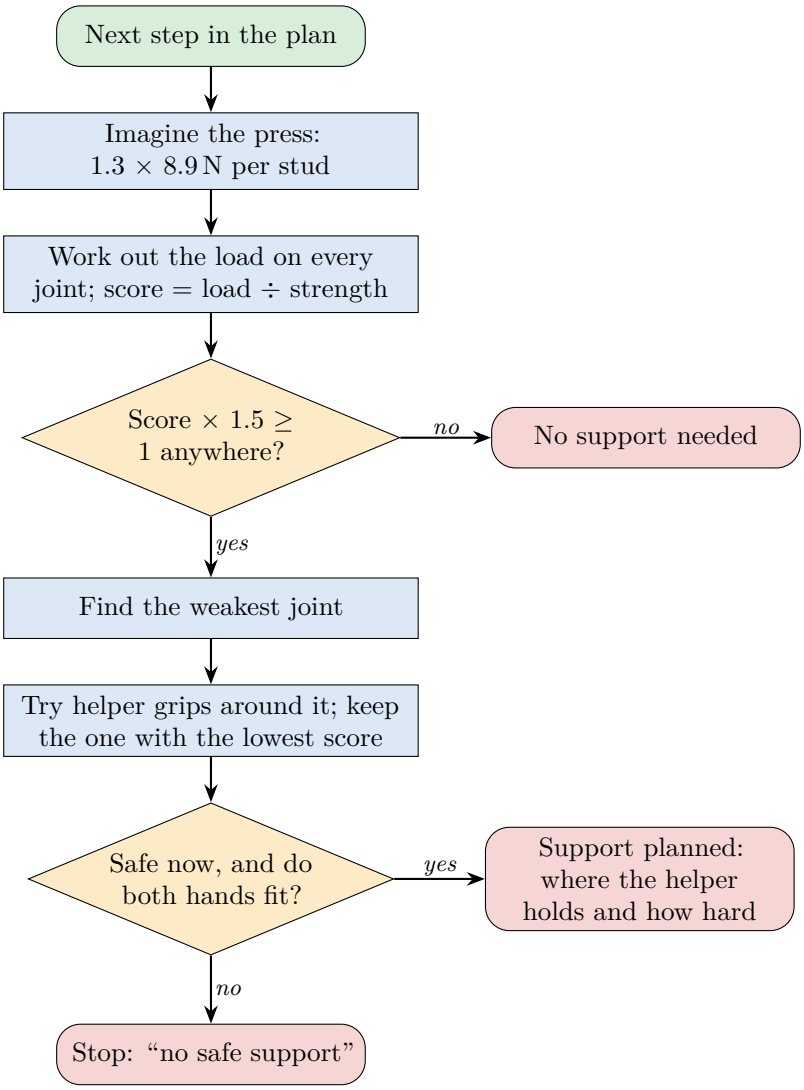


Figure 4: Deciding whether a step needs the other arm to hold the model.

5 The pretend world: how bricks are simulated

The project does not use real robots yet. It runs a **physics simulation**: a program that moves everything the way gravity, pushes and bumps would move it in real life. It runs on the computer's graphics card, which is very fast at doing many small sums at once.

- **Time in small steps.** The world moves forward 60 times a second, and each of those is split into 16 even smaller steps, so fast bumps are not missed.
- **Two shapes per brick.** Each brick has a detailed shape with studs (what you see and what the fingers touch), plus a few hidden simple blocks: a top slab, four walls and one cylinder per stud. The computer uses the hidden blocks to work out when bricks touch, because simple shapes are quick and reliable.
- **The click.** When a brick is in the right spot (see Section 6.6), invisible “glue” joints switch on between it and the bricks under it. From then on it holds like a clicked-in brick.
- **Breaking.** Every glue joint keeps checking how hard it is being pulled or bent. If that is more than the studs can hold, two checks in a row, the joint breaks and the build stops, just as a real brick would pop off.

- **The robots.** Two Franka robot arms with two-finger grippers. Each joint of the arm is driven like a stiff spring toward the angle it is told.

Two versions of the studs. In the normal version the hidden stud cylinders leave a small gap (0.6 mm each side), so a brick slides onto the studs by its own weight and the "click" is the check plus the glue. The **stud-fit version** (switched on with `-stud-fit`) is more like real LEGO. A brick stops on top of the studs and only goes down when it is pushed with 8.9 N per stud. The arm has to press it in, using a stronger, stiffer press from above.

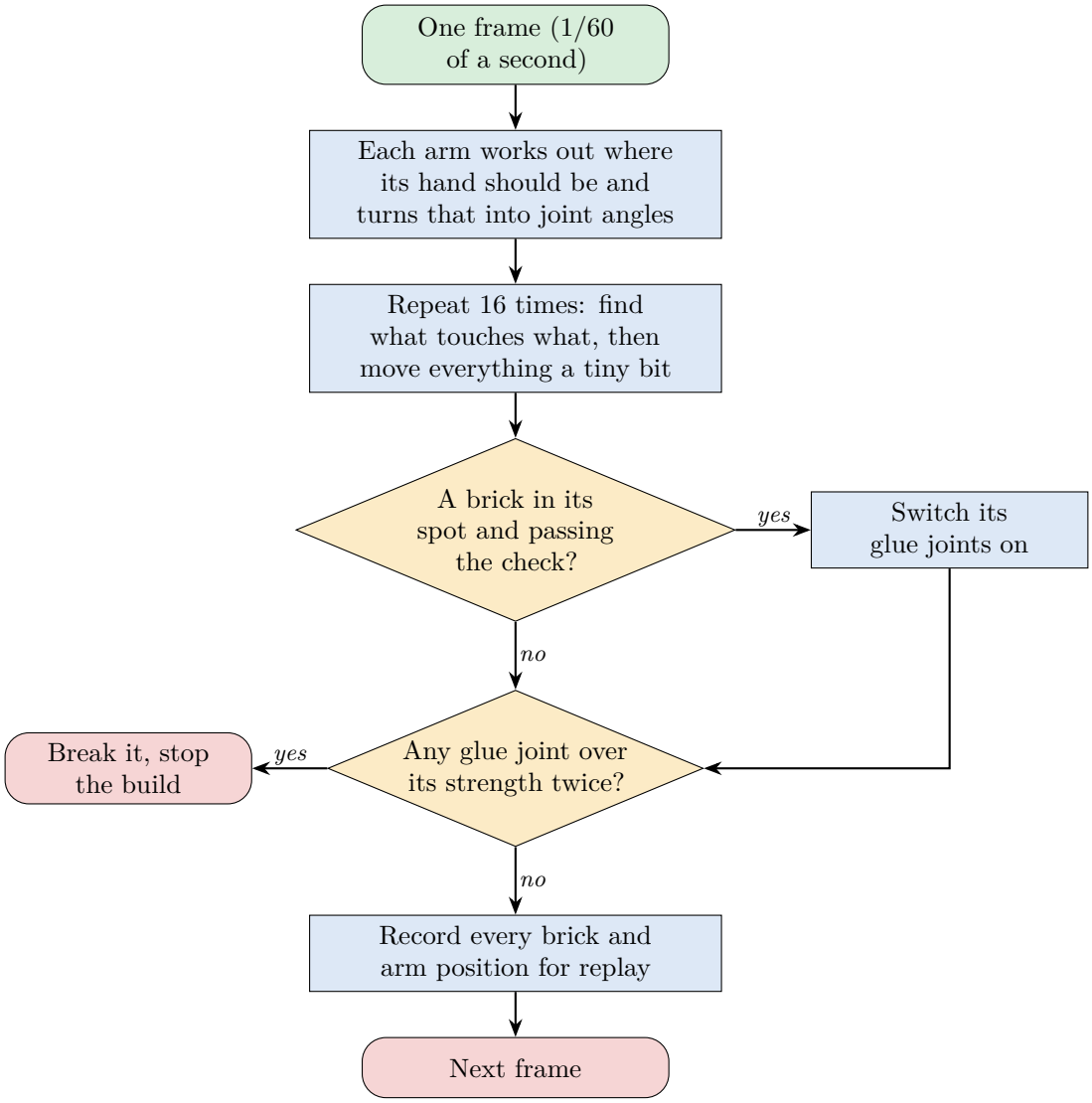


Figure 5: What the pretend world does every frame.

6 Control: how the arms do the work

6.1 Picking up and placing a brick

Each brick waits in its own spot on the table, the "feeder". To place it, the arm follows a fixed list of moves:

1. **Go to the feeder** and stop high above the brick.
2. **Go down**, close the fingers on the chosen sides, and **lift**.

3. **Carry** the brick high above everything, to the hover point over its spot.
4. **Go down slowly** toward the spot. On the way the arm watches where the *brick* really is, not just its own hand, and nudges itself by up to 4 mm so the brick lands exactly over the studs. (A brick can slip a little in the fingers, so the brick is what matters.)
5. If the spot is a tight slot, **wiggle** it gently (small side-to-side and rocking moves) so it slides in.
6. **Check the brick** (Section 6.6). If it passes, it clicks in.
7. **Open the fingers** while rising a little, **lift away**, and go to a safe resting pose ("park").

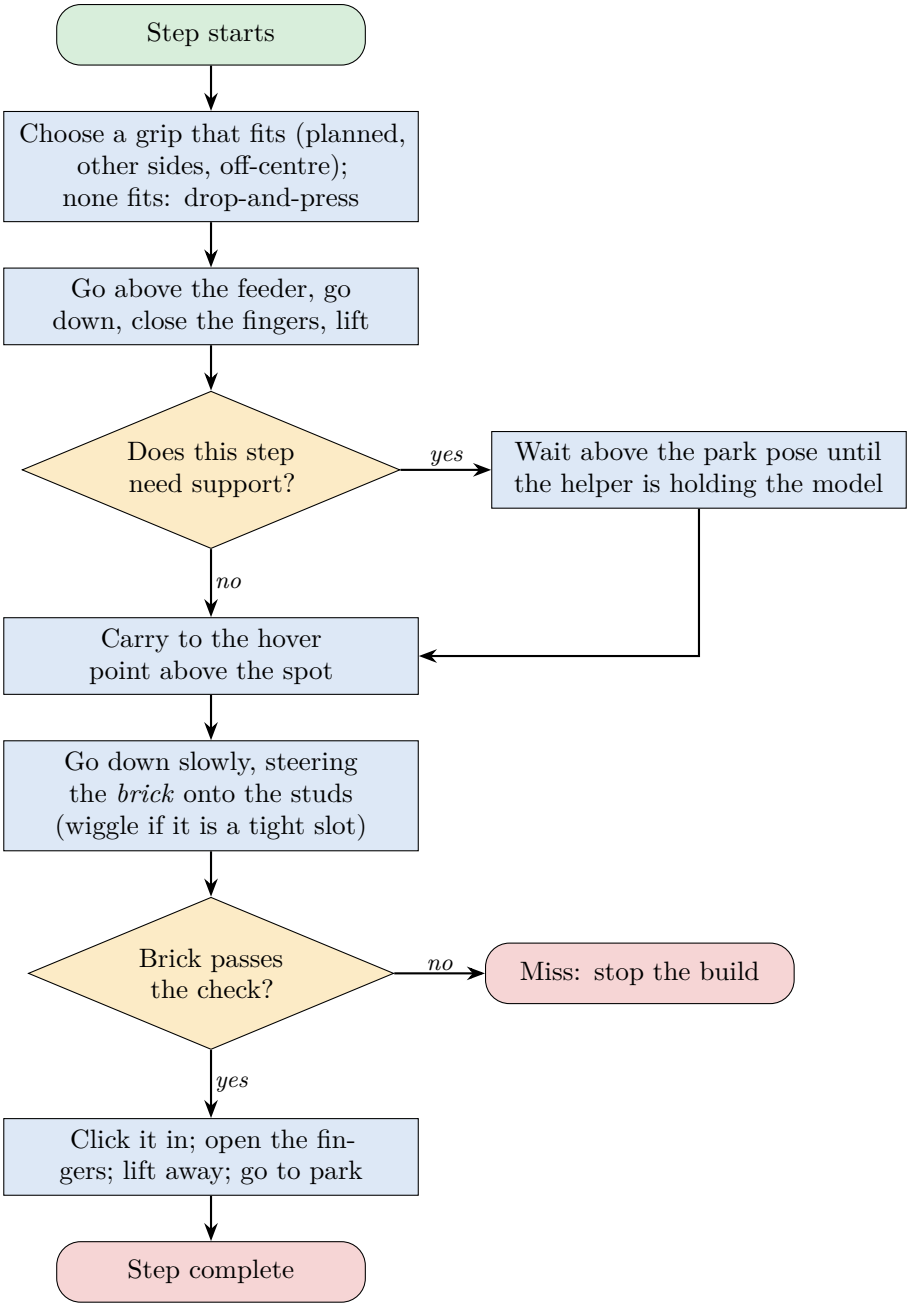


Figure 6: Pick and place, one brick.

6.2 Drop-and-press and the stud-fit press

When the fingers cannot reach down beside a brick, the arm lowers it only until the fingers are still clear of the neighbours, waits a moment, and lets go. The brick drops onto the studs. Then the arm closes its fingers into a "fist", comes down on top of the brick, and pushes it into place.

In the stud-fit version every brick is pressed this way, because the fingers can only squeeze gently (the brick would slip at about 3 N). The press is careful:

- the push grows smoothly to 1.3 times what the studs need;
- the arm (and a helping arm, if any) goes stiff while pressing, so the hand does not slide off;
- it presses where the plan says, not where an old guess was;
- the push stops the moment the brick goes in. The computer then waits a quarter of a second for the model to settle before checking it.

6.3 Finding a path: route planning

Moving a robot arm is like steering a long, bendy snake. To get from one place to another without hitting anything, the arm uses a **route planner** (called cuRobo). You can think of it as a satnav for the arm's seven joints.

- The planner is told about everything in the way: the two models, the base plates, the table, the bricks waiting in the feeders, and the other arm. Each of these is a simple box shape.
- It searches for a smooth set of joint movements that keeps every part of the arm, and the brick it carries, away from all of those boxes.
- For the short, careful moves at the end (going down onto the studs, letting go, lifting off) the arm moves its hand in a straight line instead.
- If no route is found while the other arm is busy, it waits and tries again.

6.4 Not bumping into things: safety bubbles

The computer pictures each arm as a chain of **bubbles** (spheres) that cover it like bubble wrap, and every obstacle as a box.

- **Every command is checked.** Before an arm is told to move, the computer checks the bubbles at the new position. Every bubble must stay at least 1 mm from every box, except the ones it is meant to touch: the brick in its own hand, and the bricks right next to the spot while pressing. Anything closer stops that arm.
- **The two arms share the space by booking it.** Before an arm starts a move, it "books" the space its bubbles will sweep through. If that space overlaps space the other arm has booked, it waits until the other arm is done. If both ask at the very same moment, arm A goes first and B waits its turn.
- **Staying inside the booking.** For the short straight-line moves, the arm books a little extra room around its hand. If the hand ever leaves its booked space, the arm stops.
- **Touch checks.** Every contact in the pretend world is sorted into "meant to happen" (fingers on the held brick, a brick on its studs) or "not meant to happen". The second kind fails the final check.

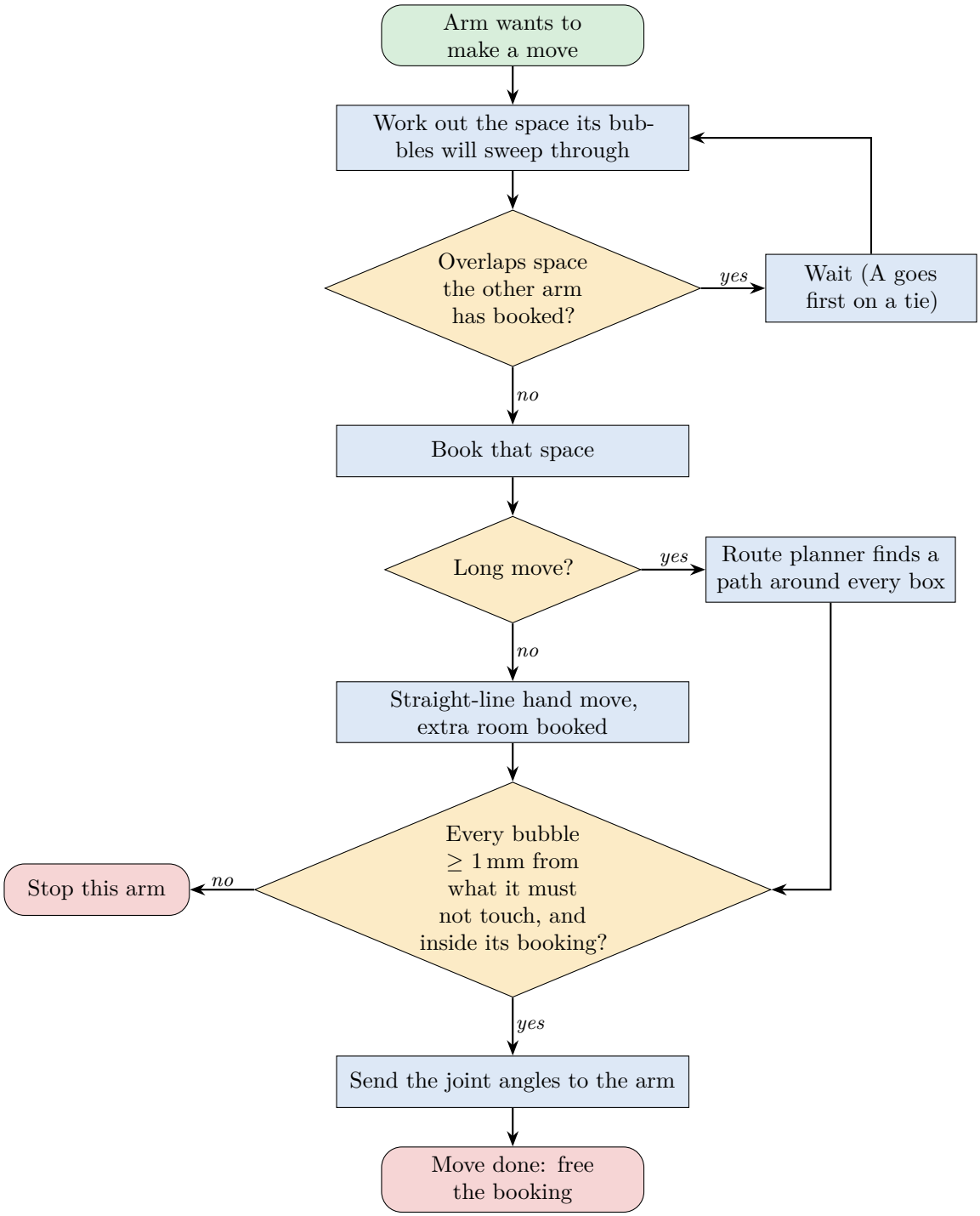


Figure 7: Planning a move and keeping it safe. The safety check runs again for every command, many times a second.

6.5 Two arms, two models, and helping each other

A small "team leader" program (the coordinator) keeps track of both builds:

- Each arm works through its own plan. When their moves do not overlap, both arms move at the same time.
- When a step needs support, the building arm asks for help. The other arm finishes what it is doing, comes over, grips the model where the support plan says, and holds it. Only then does the builder carry its brick in.

- When the brick is clicked in and the builder's hand is away, the helper lets go and returns to its own model.
- If both arms ask for help at once, A is helped first and B's request is kept for next.

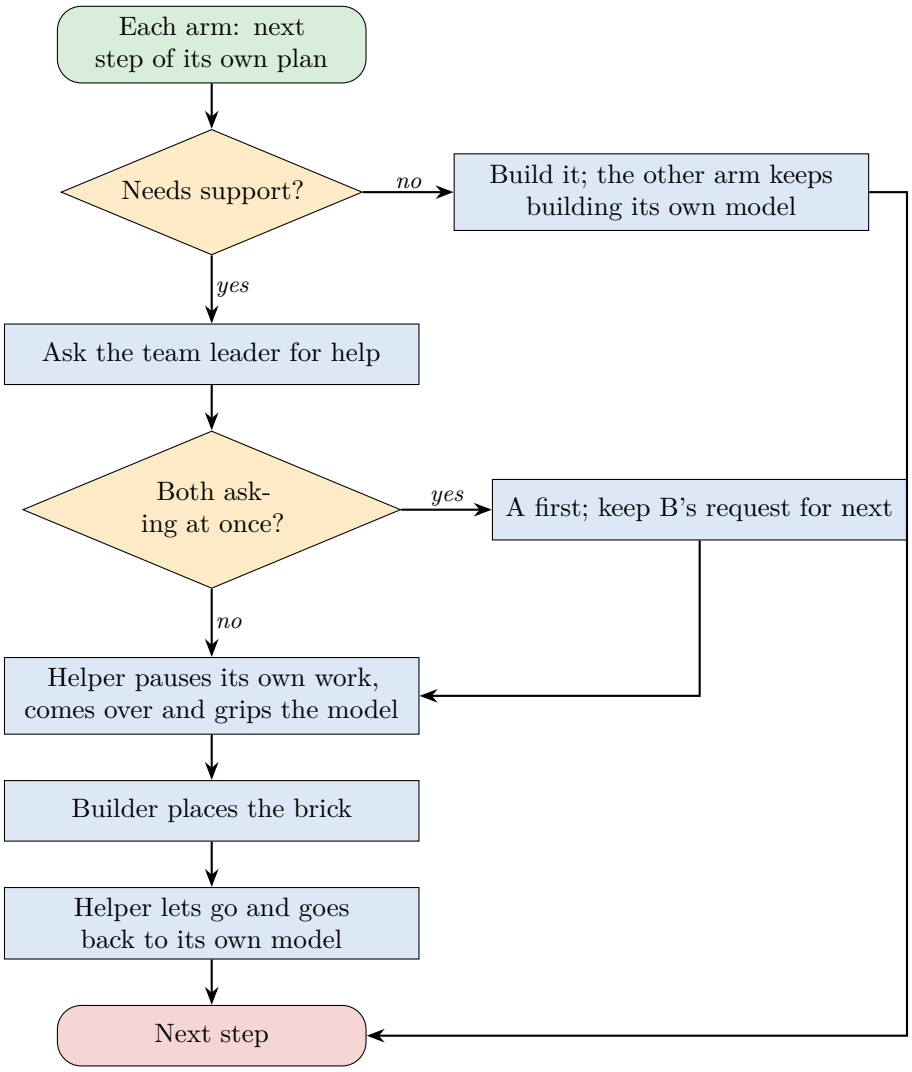


Figure 8: How the two arms share work and help each other.

6.6 When is a step finished?

A brick only counts as placed if it passes a strict check, called the **snap check**, at the moment it should click in. The computer compares where the brick really is with where the plan says it should be:

What is measured	Must be less than
Sideways distance from its spot	2 mm
Height above or below its spot	1.5 mm
Turned the wrong way	5 degrees
Tipped over	5 degrees

If all four pass, the glue joints switch on and the step is complete. The arm lets go, the team leader moves that arm on to its next brick, and any helper is released. If any one fails, it is a **miss** and the build stops, so a mistake is never hidden.

After the whole run, a separate **final check** reads the saved record of the run. It confirms that every brick of both models was placed and is still clicked in, that every support was really held, that no contact happened that should not have, and that the code that ran is the code that was saved.

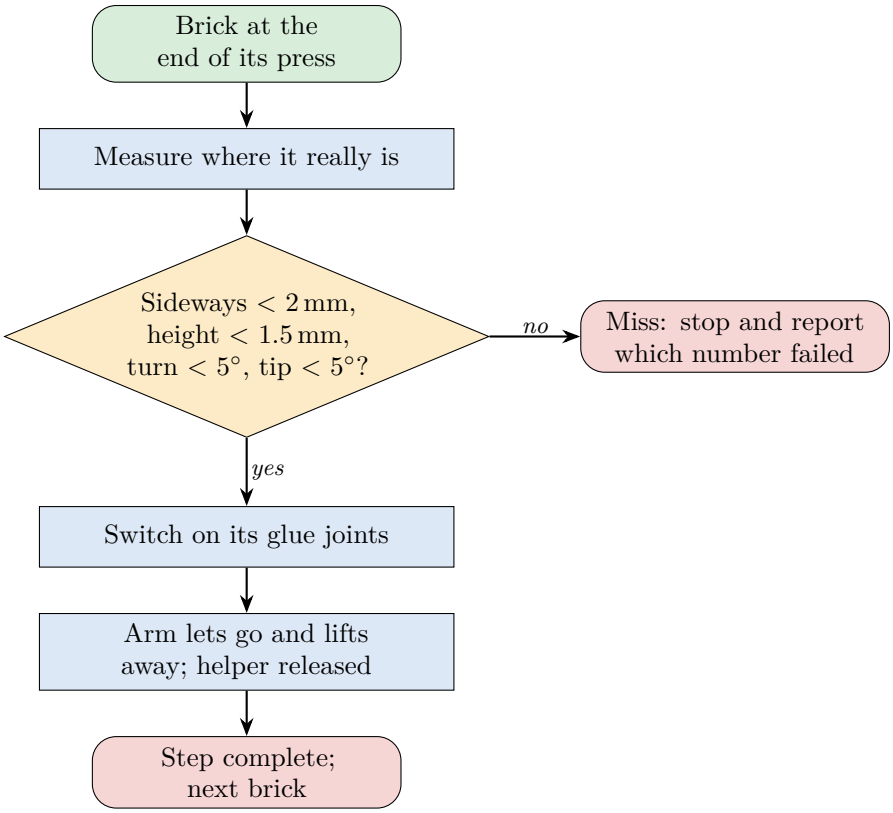


Figure 9: Deciding that a step is finished.

7 What has been achieved so far

Run	Result
Arch (arm A) and S3 tower (arm B)	Both finished, final check passed, both arms often moving at once (`dual_arch_s3_012`, `014`, `015`).
Pyramid (A) and S3 (B)	Both finished, final check passed, after reordering the pyramid's small bricks (`dual_pyramid_s3_006`).
Pyramid and S3 with real stud push (stud-fit)	The whole pyramid is built against the real stud push. S3 is built up to its second-to-last brick. On its last brick the press overloads one small 2-stud joint by a few percent, so that run does not pass yet.

Videos of the runs are saved next to their results as `.mp4` files, and every run can be replayed in a web browser.

8 Words used in this report

Blueprint The list of bricks that makes the model: type, position and direction.

Plan The blueprint put in build order, with how to grip and press each brick.

Support The other arm holding the model steady while a brick is pushed on.

Simulation A computer program that acts like the real world.

Newton (N) A unit of push; about the weight of a small apple.

Feeder The spot on the table where a brick waits to be picked up.

Hover point The spot just above a brick's place, where the arm pauses.

Route planner The program that finds a path for the arm around obstacles.

Booking Space an arm reserves before it moves, so the other arm keeps out.

Snap check The test that decides a brick is in the right place.

Glue joint The invisible link that holds a placed brick, and can break.